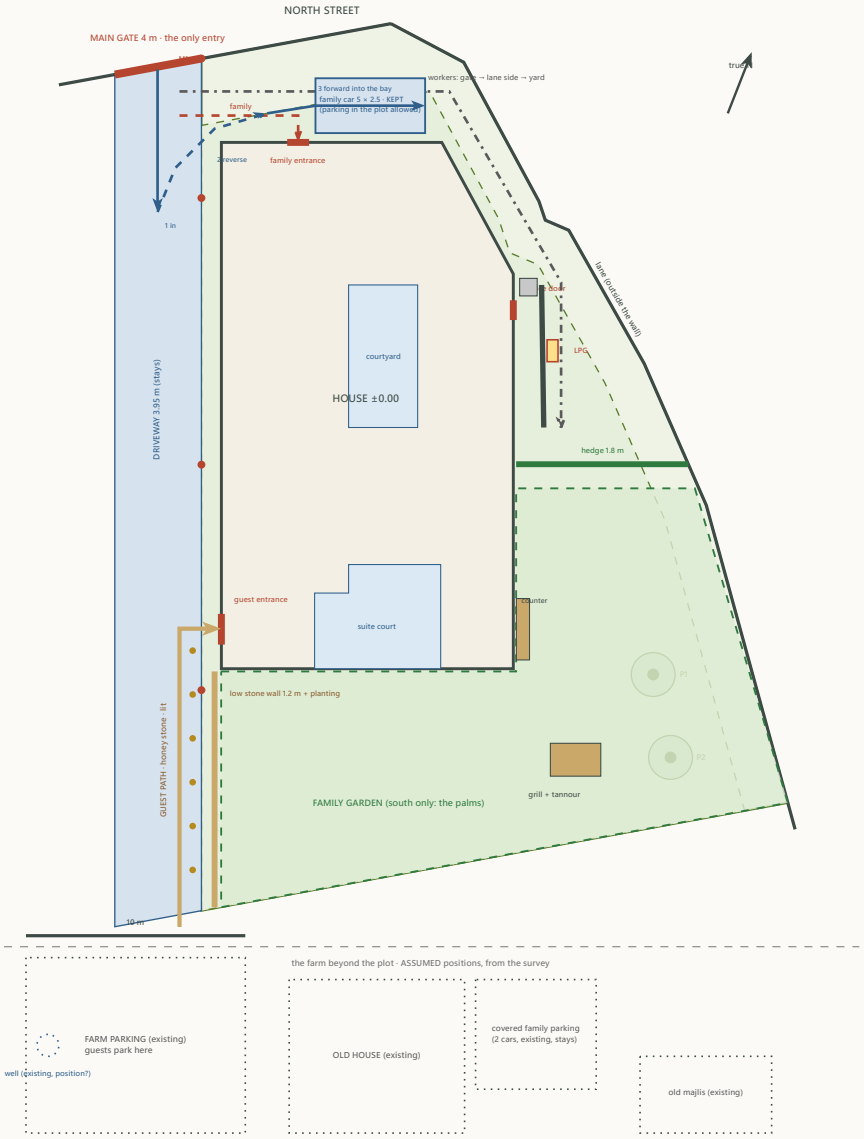
Site and services plan · access, parking, paths, the family garden, the service yard, utilities, security, Civil Defense access, levels · Taif / Palm House V8.1 · STUDY

Design review S-01 … S-08, E-04 … E-06 · open item 20 · 2 Nov 2026 · drawn on `base.typ` (the plot, the M line, the gate, the palms); the old house, the old majlis, the well, the existing parking, the existing tank and the septic are not in the project files and are drawn dotted where the owner's brief places them · **every site position is PROPOSED, to confirm with the official survey and the engineers**



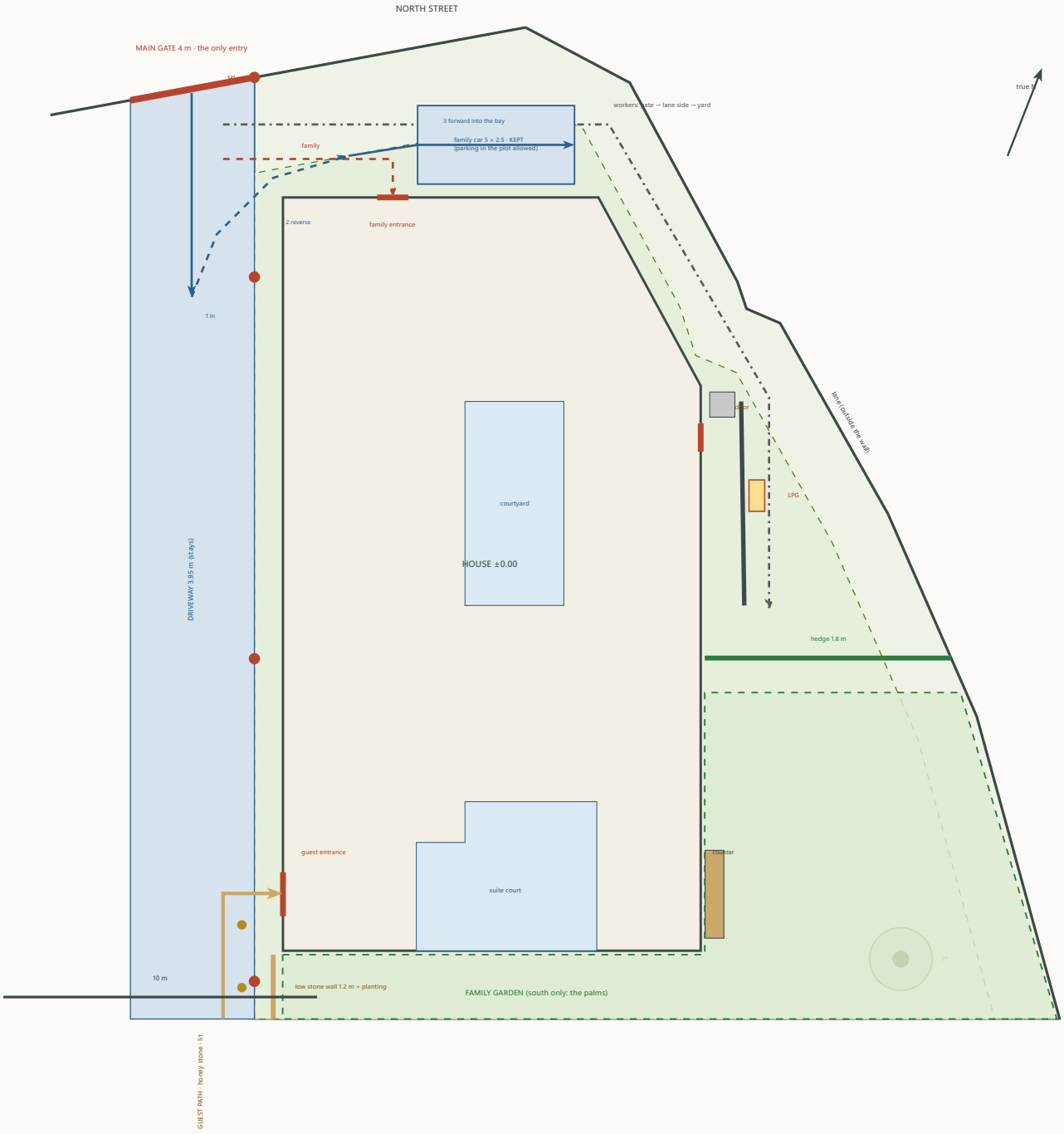
1 · Access and parking

- **One entry:** the farm is fully walled; the existing **4 m main gate** on the north street (peg M1 is its east post) is the only entry; no other gate (the lane-side gate of V8.0 is dropped). A **video intercom and a camera** at the gate (page 5).
- **The driveway** (3.95 m, west of the M line) stays: the cars' lane from the gate to the farm parking and the family's existing **covered parking for two cars south of the old house** (beyond the plot, assumed position: from the survey). No new carport.
- **One uncovered family car KEPT** (owner, 2 Nov 2026: parking inside the plot is allowed by the municipality): a 5 × 2.5 bay in the north setback strip, 0.4 m off the house's north wall, 0.29 m east of the family entrance's path, which stays clear. **The swept path** of a standard car (4.5 × 1.8, outer radius ≈ 5.5 m) shows that a forward 90° turn from the driveway into a 4–4.6 m deep strip does not fit; the car enters in a **3-point manoeuvre** (in through the gate and 7 m south on the driveway; reverse north-east into the strip; forward west into the bay, nose west) and leaves forward onto the driveway; the parked car stays clear of the entrance, and the manoeuvre crosses the entrance path at grade (the kerb flush there), as on any driveway. **Improvement if wanted:** round the family entrance path's kerb and keep the reserve tank's lid flush, so the strip's west end (x 0–5.2) is usable as the turning pocket; a shorter car fits in one move.
- **Guests** park in the large existing **farm parking south of the building** (no new design) and walk the **guest path**: a 1.2 m **honey-stone band along the driveway's east edge** from the plot's south end north to the **guest entrance (the west foyer)**, 15.5 m, with bollard lights every 2 m; it never enters the family areas (the garden is behind its low wall, the north strip and the east yard are far). The driveway keeps 2.75 m beside it for one car.
- **The family** enters by the north: gate → the north strip → the **family entrance** (6.6 m); **workers and deliveries**: gate → the north strip → the lane-side strip (2.5–3 m wide east of the cut wall) → the **service yard** east of the screen (29.2 m), never through the garden or past the guest door.

2 · The family garden and its separation

- **A family garden on the south side only**, round the two palms, from the south facade to the plot's south line (≈ 289 m²); **no other garden** round the house: the north strip is the entrance and the optional car, the west strip the driveway, the east strip the service side.
- **Separation:** on the west a **low honey-stone wall 1.2 m with planting behind it** between the garden and the guest path on the driveway (a wall, not a hedge: it also holds the garden's soil 0.3 m above the drive and carries the path's lights); on the north-east a **1.8 m hedge** from the house's east wall to the boundary at y 18.5 between the service yard / workers' path and the garden (a hedge, not a wall: the yard's air and the lane-side breeze pass). The grill and tannour and the outdoor counter are inside the garden side.
- **No garden door from the living** (review S-04: **not needed, owner**): the **kitchen service door** is the family's door to the outside on the east; the garden is reached from it round the south-east corner (the hedge has a gate there) or from the guest entrance side.

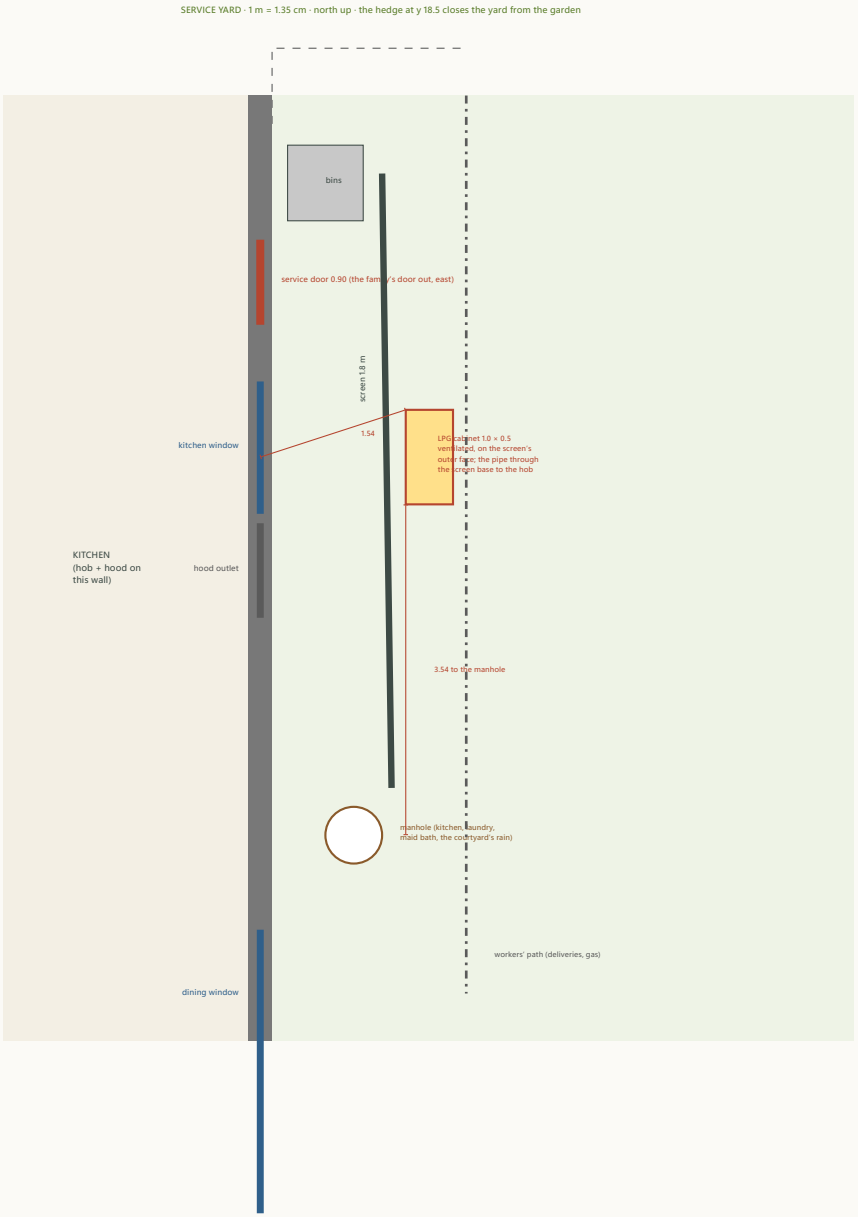
Around the house (enlarged, y ≤ 30 m) · the strips, the paths, the family car (kept), the garden's edges



Around the house

- **North strip** (4–4.6 m to the boundary): the family entrance path from the driveway, the family car (kept), the reserve for the future tank (page 4), the workers' path along its north edge. **West strip** (0.785 m inside the M line) + the driveway: the guest path, the rescue ladder stand for bedrooms 2 and 3 (fire study P2: keep it open at night), the fire vehicle. **East strip** (2.5–9 m to the lane): the yard, the workers' path, the rain soakaway. **South:** the garden.
- **Levels (page 4):** the house floor ±0.00 is assumed 0.30 m above the ground around it; every strip falls 2 % away from the walls; the driveway and the north strip fall to the gate; **to be set from the survey** (the street level at the gate, the farm's levels, the lane).

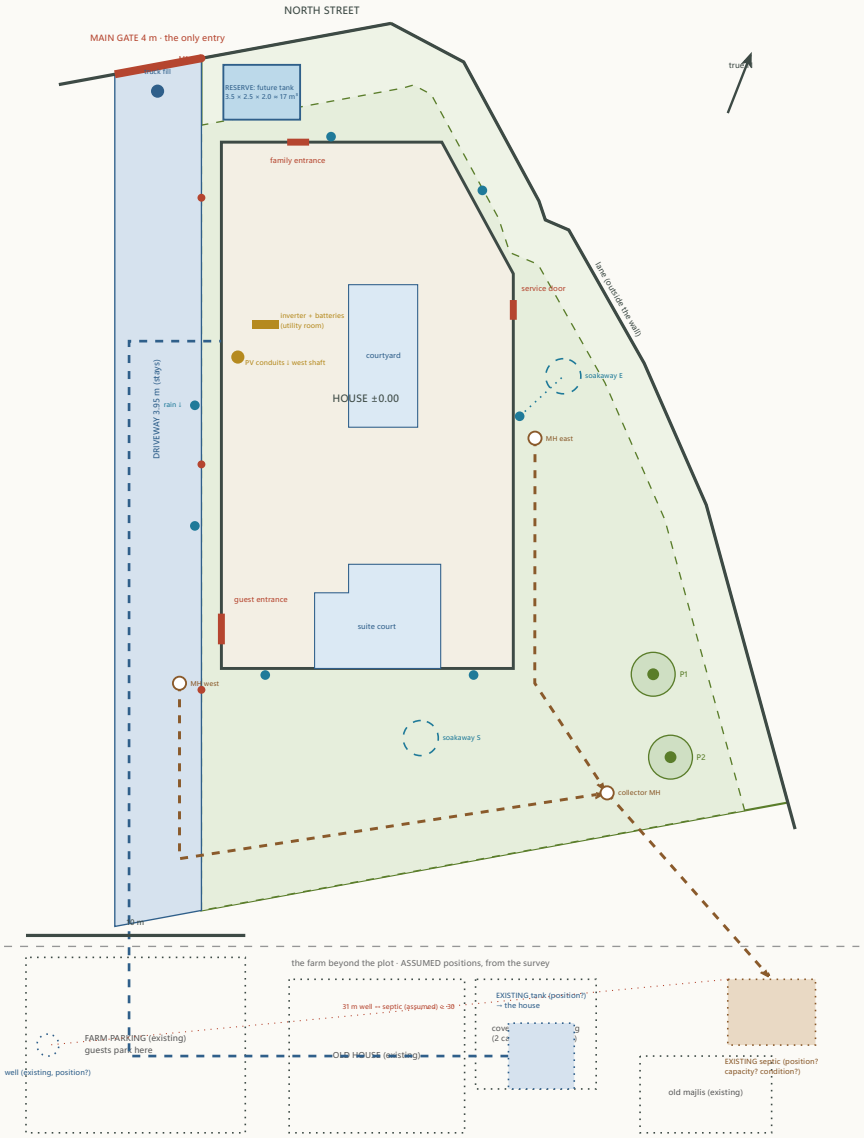
The service yard (enlarged): the gas cabinet, the bins, the manhole, the screen, the workers' path · the rule used for the gas cylinders



3 · The service yard (east)

- **At the kitchen service door** (the family's door out): the **bins** north of the door against the wall, the **1.8 m screen** 1.3 m from the wall closing the yard from the lane side, the **manhole** at the yard's south end. The **outdoor kitchen** as drawn: the counter and sink against the east wall further south (y 20.8–23.6 in the CSV frame, in the house's afternoon shade; a canopy proposed), the grill and tannour apart in the garden (3.6 m from the house's corner, 3.2 m from the nearest palm).
- **The gas cylinders:** in a **ventilated outdoor cabinet 1.0 × 0.5 on the screen's outer (east) face** (CSV x 14.85–15.35, y 9.0–10.0), so the delivery man changes cylinders from the workers' path without entering the yard; the pipe through the screen's base to the hob on the east wall (≈ 3 m). **The rule used (assumed, to confirm with the Civil Defense):** NFPA 58 as adopted in Saudi practice for small cylinders outdoors: ≥ **1.5 m** from any opening into the building below the cylinder's relief valve (doors, windows, the hood outlet), ≥ **3 m** from ignition sources, mechanical air intakes, drains and manholes (LPG is heavier than air), the cabinet ventilated top and bottom, the cylinders upright, the regulator and the shut-off outside, the gas detector and the automatic valve inside (fire study). **Measured:** the service door 1.78 m, the kitchen window 1.54 m, the hood outlet 1.55 m, the dining window 4.76 m, the manhole 3.54 m, the grill 17.4 m, the HRV intake on the roof > 10 m, the lane boundary 2.9 m. **OK**
- **Screened from the guests' path:** the yard is on the east, the guest path on the west; nothing of the yard is seen from it. The **workers' path** runs along the lane side east of the screen and ends at the yard; the **hedge** at y 18.5 stops it short of the garden.

Utilities · reserve positions and routes, no system design · water, sewage, power (PV only, no generator), rainwater · levels



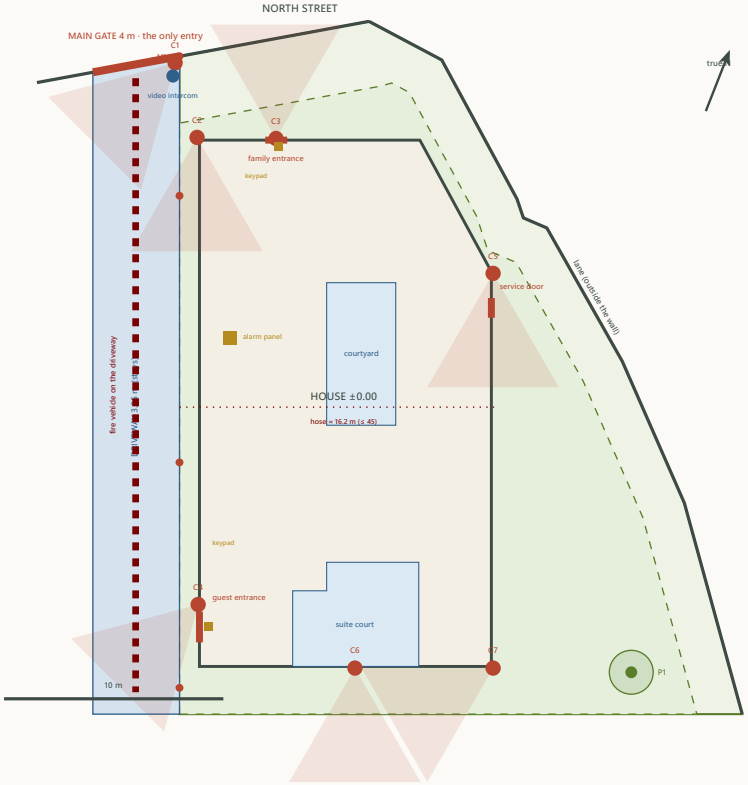
4 · Utilities

- **Water:** the **existing ground tank stays** (position assumed near the old house: from the survey); its supply pipe to the house is drawn along the driveway's west edge to the **utility room** (the manifold, the heater, the main valve) through the west wall: 54.1 m assumed; the engineer checks the pump and the pipe size. **Reserve for a future new ground tank** near the house: an underground tank 3.5 × 2.5 × 2.0 m (≈ 17 m³, a 20 m³ class) under the north strip beside the family entrance path (CSV-frame x 0.1–3.6, y −3.5 … −1.0), 1.0 m off the house wall, with its **truck fill point on the driveway** 2 m away (the only vehicle access); the municipality's water truck stands inside the gate. Also the fire reserve, if the Civil Defense asks for one.
- **Sewage:** the house connects to the **existing septic tank** (position assumed south-east beyond the plot: from the survey): the east manhole (the kitchen, the laundry, the maid bath, the suite stack) runs south along the east strip; the west manhole on the driveway (the west wash block's run) runs south and east under the garden's south edge; both meet a **collector manhole** at the garden's south-east, then ≈ 11.3 m to the septic. Falls 1–2 % (≈ 0.5 m over the run): the septic's inlet must lie ≥ 1.0 m below the house's ground floor. **Flags for the engineer:** the septic's capacity for the new house (≈ 12 persons with staff), its condition and age, its distance from the well (≥ 30 m in Saudi municipal practice; drawn 31 m on assumed positions), the soakaway's soil (the soil report, open item 7).
- **Power: solar PV provision only** (the roof zones reserved: Z4a / Z4b on the south wing, Z5 on the north block); the **conduits** from the roof come down the **west shaft** to the **utility room**, where the **inverter and the battery cabinet** go (owner, 2 Nov 2026: the utility room; the electrical engineer confirms the space, the ventilation and the fire safety; the room's layout with the water kept away from the batteries is in `studies/plumbing.typ`). **No generator** (review E-05: **no generator, owner**). **Note:** until the PV and the batteries are installed there is **no backup during power cuts** (the lift, the HRV, the pumps and the gas valve stop; the gas valve fails closed).
- **Rainwater:** the roof's 11 outlets reach the ground in internal downpipes and discharge at the plinth: north (1), west (2, into the driveway's channel), the cut and the east wall (2, into the **east soakaway** in the strip), south (2, into the garden); the **courtyard's drain and overflow** and the **suite court's gutter** go east to the manhole's **rain branch** and the east soakaway, **never into the septic**; a second soakaway in the garden takes the south downpipes. The soakaways are sized by the soil report.

7 · Levels (to be set from the survey)

- **Assumed:** the house floor ±0.00; the ground around the house at −0.30 (the plinth 0.6 m shows 0.3 above it); the courtyard's seating −0.15 inside; every strip falls 2 % away from the walls for 2 m, then towards the soakaways; the driveway and the north strip fall gently to the gate (the street level unknown); the garden at −0.30 … −0.60 with the palms' existing ground kept; the yard at −0.30 with its manhole; the farm parking and the old house: unknown.
- **From the survey:** the street level at the gate, the plot's levels at the pegs M1–M4, the lane's level along the east boundary, the palms' trunk bases, the old house's floor, the septic's inlet level, the well's head. Then the engineer sets the finished floor against the street and the drainage falls.

Security · Civil Defense access · the items for the survey, the municipality and the engineers



5 · Security

- **The main gate:** a video intercom (the only entry) and camera C1 on the gate post looking at the gate and the street; the gate's motor and the intercom on the house's alarm circuit.
- **Cameras round the house** (black fittings, high on the walls under the band or the cornice, IR at night): **C1** the gate (with the video intercom); **C2** the driveway, looking south; **C3** the family entrance; **C4** the guest entrance and the path; **C5** the service yard, looking south; **C6** the garden and the south facade; **C7** the suite court's outside (the screen) and the east garden. Together they cover the gate, the driveway, the family entrance, the guest entrance, the service yard, the garden and the suite court's outside (its mashrabiya screen and the south-east garden). The recorder with the **alarm panel in the utility room**; keypads at the family entrance and the guest entrance; contacts on the three exit doors and the suite court's doors; the gas detector and the smoke alarms on the same panel (fire study). **The electrical engineer** designs it; positions proposed.

6 · Civil Defense access

- **Can a fire vehicle reach the house?** The gate is 4.02 m (≥ 3.5 **OK**), the driveway 3.95 m wide and straight for 39 m (≥ 3.5 **OK**): a vehicle stands on the driveway along the whole west facade, 0.8 m from it, and sees the north facade from the street and the gate. **There is no turning area inside the plot**: the vehicle turns in the farm parking south of the plot (to confirm that it is reachable and large enough) or reverses out. **Hose reach:** from the driveway round the house to the east wall ≈ 16.2 m, the farthest point of the house (≤ 45 m **OK**). The east facade is also reached from the **lane** outside the farm wall (2.5–9 m away) if the lane is drivable (to confirm), the south facade from the farm parking.
- **The engineering office must confirm** (SBC 201 / the Civil Defense's villa checklist): the required approach distance to the facades, the hose reach, the water supply (the tank as a fire reserve, a hydrant on the street?), the gate's clear height, and the rescue ladder stand on the driveway for bedrooms 2 and 3 (fire study P2). **to confirm**

Items for the survey

1. The old house, its covered parking, the farm parking, the old majlis, the well, the existing ground tank, the septic tank and the farm's walls: their positions and levels (all assumed here).
2. The street level at the gate, the levels at the pegs, the lane's level, the plot's south line, the palms' trunks.
3. The legal boundaries and the setback lines (open item 1): the north street line and the lane line against the envelope used here.

Items for the municipality

1. Parking in the north setback strip (one uncovered car): allowed?
2. The septic tank's reuse for the new house; the rain soakaways; the discharge to the lane side.
3. The gas cabinet and the Civil Defense's villa requirements (access, water, alarms).

Items for the engineers

1. Civil / MEP: the septic's capacity and condition, the sewer falls, the well distance; the water tank's pump and pipe; the new tank's size and the fill point; the soakaways from the soil report; the PV conduits and the inverter / battery space; the gas pipe and the cabinet's details; the levels and the drainage round the house.
2. Electrical: the cameras, the intercom, the alarm panel, the gate motor, the night lights on bedroom 1's route, the outdoor lighting of the paths.

Study checks passed (10)

Checks: the gas cabinet ≥ 1.5 m from every opening of the house (service door 1.78, kitchen window 1.54, hood outlet 1.55, dining window 4.76) · the gas cabinet ≥ 3 m from the manhole (3.54) and the grill (17.4) · the gas cabinet ≥ 1.5 m from the lane boundary (2.9) · the family parking bay (kept) inside the plot, 0.4 m off the house's north wall, 0.29 m clear of the family entrance's path · the guest path never enters the family garden (west of its low wall) and ends at the guest door · the workers' path stays north of the garden's hedge and east of the house · the new tank's reserve inside the plot and ≥ 1.0 m from the house's north wall · the gate (4.02 m) and the driveway (3.95 m) take a fire vehicle (≥ 3.5 m); every point of the house within 45 m of hose from the driveway (≈ 16.2 m) · the grill ≥ 3 m from the palms (3.2)) and the gas cabinet (17.4) · the assumed septic ≥ 30 m from the assumed well (31): both positions from the survey. Every position is proposed, to confirm with the official survey and the engineers; the items outside the plot are assumed.